

Production testing

After the silicon has passed the tests and the chip is sent for mass production, every chip produced undergoes similar tests. Every chip is tested for its functionality and characterisation. Testers are preloaded with test programs to test the ICs. There are tools for signal integrity and thermal management tests.

A go-no-go test is performed on every chip in big test assemblies called ATEs or testers. Veerapan explains, "There are test load boards to load the chip into the tester. The test program is customised. Every pin of the IC has to be activated by some signal, and the tester will send the signal and receive an output, which will then be compared with the expected output. So, basically, the tester simulates the test as if the end customer is using the chip."

Designing of test load boards is another critical area requiring skilled test engineers. Varambally adds, "Another key consideration is that one tester can test only one IC or range of ICs of a particular category. There is no universal tester that can be used to test all the ICs in the world."

For better and faster testing

The design and test methodology has to evolve with the increasing complexity of silicon design. "Testing is becoming an integral part of design. Unless you have very good test methodology built into your design tool and at design stage, it will lead to critical problems. So designers and test engineers have to work hand-in-hand. Unless these guys work together, testability will become a problem later," claims Bhardwaj.

Concurrent testing is an effective method to drive the test times. This can be accomplished by testing different cells of the same silicon in parallel, i.e., testing multiple cells at the same time wherever possible. As the SoC is a combination of various digital cores like ADCs, DACs and

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digital and analogue cells, all of these can be tested in parallel and then the entire device tested for functionality. This will reduce the overall time significantly.

"Another test strategy offered by ATE manufacturers is parallel multi-site testing where multiple devices can be tested simultaneously (at more than 128 sites), reducing the overall test time and increasing throughput significantly," comments V.K. Rajiv, product manager, Qmax.

Many semiconductor manufacturers like Intel and TI have home-grown ATEs built using in-house tools. They have their own test programs, and only the conversion tools are provided by EDA vendors.

Veerapan explains, "There is no common test program which can test all the chips. You need to customise the program to your chip. Test engineers have to develop the right skills to write efficient test programs."

Rajiv adds, "Many ATE manufacturers now provide test program development software that are fully graphical user interface based and also automate many development procedures to facilitate faster test program development and minimise programming errors."

Design of the load board is another critical area. Veerapan shares, "It has to have all the precautions inbuilt. For example, in case of high voltage, it should have a high-voltage cut-off. All types of arrangements are made in the circuit of the load board."

Test program reuse can significantly reduce the cost and time of testing. Veerapan says, "Reuse is a smart way of reducing time. Currently, 30-40 per cent of the test programs, boards and tools can be reused. Learning of the re-

use technique saves a lot of time spent in re-inventing a wheel. It reduces the time-to-market."

"With rapidly evolving semiconductor technologies, chips go from introduction to obsolescence in less than three years. Obviously, time-to-market and manufacturing cost of new products are extremely important to the company's bottom line, and one key element that drives both these factors is semiconductor testing. ATE manufacturers need to ensure that the existing ATEs are highly scalable and adaptable to the rapidly changing semiconductor industry," adds Rajiv.

Ultimate goal: minimise cost and time, maximise yield

The operating speed of ICs has increased greatly. The test systems should be able to match the speed of these faster chips. Another area of development is the manufacturing process technology. As manufacturing moves from 90nm to 65nm process, the tester should be capable of handling the lower footprint and producing reliable test results.

Process technology changes have also had serious effect on the cost of test and the need for test. Kumar adds, "With 32nm technology becoming a reality in 2010, test challenges have only increased. Sub-pico-Farad capacitance measurements and tens of atto-ampere (current) measurements have become essential."

Today, chips have become very complex incorporating about four million gates. The ATE should be able to test these chips in minimum possible time. You could spend 20 minutes or five seconds to test a chip and every second adds to the cost of chip production. The ultimate goal of any chip manufacturer is to be able to test all the chips in minimal time. This will reduce the cost and increase the yield. ●

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